20. Final Project Report

Electric Vehicle Market Performance & Insights

Abstract

This project analyzes a supplied electric-vehicle dataset using descriptive statistics and Tableau visualization. The raw CSV contains 100,000 records and 37 fields. It spans model years 2018–2026, 200 manufacturers, 15 categories, and 7 regions. The project deliverables are a documented data profile, quality audit, exploratory analysis, dashboard, story, and responsive webpage. Because source provenance and several field definitions are not provided, findings are interpreted as patterns in the supplied file rather than verified market-wide facts.

1. Introduction

The dataset includes vehicle specifications, battery and charging attributes, price and ownership measures, safety/autonomy indicators, and market/customer fields. A dashboard helps users compare the data without manually reviewing every row.

2. Objectives

- Profile the dataset and document all fields.
- Assess missingness, uniqueness, and suspicious numeric ranges.
- Explore distributions and relationships.
- Present comparisons through Tableau.
- Embed the views in a website and document limitations.

3. Dataset

The file contains 100,000 rows and 37 columns. It has 200 manufacturer labels, 10 manufacturing countries, 7 market regions, 15 categories, and 5 battery types. The model-year range is 2018–2026.

4. Data quality and preparation

There are 9,511 missing cells (0.257% of all cells), no complete duplicate rows, and no duplicate vehicle IDs. Missing values occur in `battery_cycle_life`, `resale_value_pct_3yr`, `annual_sales_units`, and `customer_rating`. Extreme values require source validation; they should not be automatically removed.

5. Methodology

The analysis uses univariate summaries, group comparisons, distributions, and Pearson correlations. Mean and median are both reported for skewed metrics. Charts should disclose units, aggregations, filters, and effective sample sizes. Correlation is interpreted as association only.

6. Results

- Mean price: $65,010.46; median price: $42,808.69.
- Mean range: 324.31 km; median range: 338.30 km.

- Mean battery capacity: 78.23 kWh; median: 63.30 kWh.
- Most frequent category: Two-Wheeler Scooter (9,826 records).
- Price/battery capacity correlation: r = 0.837.
- Price/range correlation: r = 0.277.
- Customer rating/price correlation: r = -0.001.

These are raw-file statistics. If the Tableau workbook uses filters or cleaning, the final published values may differ and must be reconciled.

7. Tableau implementation

The dashboard should combine summary KPIs, category/region composition, price and range comparisons, and technical relationship charts. The story should guide users from context through comparisons and finish with limitations. Each calculation should be validated against the raw data.

8. Website implementation

The website uses an HTML page based on a Bootstrap template and embeds the Tableau dashboard and story. Direct links should be provided as fallbacks. Responsive layout and public access must be tested in a browser before claiming deployment readiness.

9. Limitations

The CSV's source, sampling method, and data-generation process are not supplied. Counts cannot automatically be treated as market shares. The sales, CO■ savings, charging-time, cycle-life, autonomy, and customer-rating fields need authoritative definitions. Extreme values may materially affect means.

10. Conclusion

The supplied dataset supports a descriptive exploration of EV attributes and their relationships. The most defensible output is a transparent, interactive comparison of the records with clear definitions and caveats. Stronger real-world conclusions require source verification and validation of the measurement methods.

References and deliverables

- Source file: `ev_market_test.csv` (provided in the project conversation).
- Tableau dashboard: use the published dashboard URL supplied by the project author.
- Tableau story: use the published story URL supplied by the project author.
- Website: `index_ev_market.html` (rename to `index.html` when placing it in the new project folder).